

Strategic Objectives & Key Results

Operational Excellence

Achieve $\geq 99.5\%$ supply continuity; reduce Non-Revenue Water (NRW) below 15%

Water Quality & Safety

100% compliance with WHO & national drinking water standards; zero public health incidents

Infrastructure Resilience

Reduce pipe burst frequency by 30%; extend average asset lifecycle by 20%

Water Conservation

Reduce per capita consumption by 15% by 2030; expand recycled water program

Customer Centricity

CSAT $> 85\%$; first-contact resolution $> 80\%$; digital self-service adoption $> 70\%$

Digital Transformation

Full AMI deployment; smart network monitoring across 100% of the distribution system

Financial Sustainability

Cost recovery ratio ≥ 1.2 ; operating efficiency ratio $< 70\%$

Environmental Stewardship

Net-zero energy in treatment by 2035; reduce chemical usage by 20%

Business Capability Map

Customer Operations

- Customer Enrollment & Onboarding
- Billing & Payment Processing
- Customer Service & Support
- Complaint & Dispute Management
- Water Conservation Programs
- Demand-Side Management

Water Source & Treatment

- Source Water Procurement & Management
- Raw Water Quality Monitoring
- Water Treatment Operations (WTP)
- Chemical Dosing & Process Control
- Effluent Quality Compliance
- Desalination / Recycled Water Management

Distribution Network Operations

- Pumping Station Operations
- Reservoir & Storage Management
- Pressure Zone Management
- Leakage Detection & NRW Control
- Valve & Hydrant Management
- Network Monitoring & SCADA

Asset & Infrastructure Management

- Asset Lifecycle Planning
- Pipe & Network Rehabilitation

| Preventive & Corrective Maintenance

| Capital Project Delivery

| Asset Condition Assessment

| GIS Network Management

Wastewater & Environmental

| Wastewater Collection Operations

| Wastewater Treatment (WWTP)

| Sludge & Biosolids Management

| Environmental Monitoring

| Stormwater Management

| Recycled Water Production

Regulatory & Compliance

| Water Quality Regulatory Reporting

| Environmental License Management

| Health Authority Compliance

| Tariff & Rate Case Management

| Safety & Risk Compliance

Financial Management

| Financial Planning & Budgeting

| Revenue Management

| Cost Management & Optimization

| Capital Investment Planning

| Risk & Treasury Management

| Audit & Controls

Technology & Data

| IT/OT Infrastructure Management

Application Portfolio Management

Cybersecurity & OT Security

Data Management & Analytics

Digital Platform Management

Smart Metering & IoT

Core Value Streams

1. Connect Customer to Network

Application



Feasibility



Design



Construction



Inspection



Commissioning



Activation

2. Treat & Deliver Safe Water

Source Intake



Treatment



Quality Test



Storage



Pump & Distribute



Monitor



Customer Tap

3. Bill & Collect Revenue

Meter Read



Validate



Calculate



Present Bill



Process Payment



Collections



Reconcile

4. Detect & Repair Leaks

Detect Anomaly



Investigate



Isolate Zone



Dispatch Crew



Repair



Restore Supply



Verify & Close

5. Manage Wastewater

Collection



Pre-Treatment



Primary Treatment



Secondary Treatment



Quality Check



Effluent Discharge

Organization Structure

Governing Body

Board of Directors / Regulatory Authority

Executive Leadership

CEO / Managing Director

C-Suite

COO

CFO

Chief Customer Officer

CIO / CTO

Chief Regulatory Officer

CHRO

Operations Division (COO)

Water Treatment

Distribution & Network

Wastewater Operations

Asset Management

Leakage & NRW Control

Finance Division (CFO)

Financial Planning & Analysis

Revenue Management

Treasury

Accounting

Customer Division

Customer Service

Billing & Payment

Marketing & Communications

Water Conservation Programs

Technology Division (CIO/CTO)

IT Operations

OT / SCADA Systems

Data & Analytics

Cybersecurity

Regulatory & Environment Division

Water Quality & Lab

Environmental Compliance

Regulatory Affairs

Health & Safety

Human Resources Division (CHRO)

Talent Management

Safety & Training

Labor Relations

Business Process Architecture

1. Source & Treatment

1.1 Raw Water Abstraction Planning

1.2 Water Treatment Process Management

1.3 Chemical & Dosing Management

1.4 Quality Testing & Certification

2. Network Distribution

2.1 Pressure Management & Zoning

2.2 Pumping Station Operations

2.3 Storage Reservoir Management

2.4 Network Monitoring & Control

3. Customer Management

- 3.1 Service Connection Management
- 3.2 Meter Reading & Validation
- 3.3 Billing & Revenue Collection
- 3.4 Customer Enquiry & Complaints

4. Asset Management

- 4.1 Asset Register & GIS Maintenance
- 4.2 Planned & Preventive Maintenance
- 4.3 Emergency Repair & Response
- 4.4 Capital Works Program Delivery

5. Wastewater Management

- 5.1 Sewer Network Operations
- 5.2 Wastewater Treatment Process
- 5.3 Biosolids & Sludge Management
- 5.4 Effluent Quality Monitoring

6. NRW & Leakage Control

- 6.1 District Metered Area (DMA) Analysis
- 6.2 Leak Detection & Survey
- 6.3 Meter Accuracy Management
- 6.4 Illegal Connection Investigation

Core Data Domains & Key Entities

Customer Domain

- Customer
- Account
- Premise / Property
- Service Point

Service Agreement

Tariff Classification

Asset Domain

Treatment Plant

Pipe Segment

Pump & Pumping Station

Reservoir / Storage Tank

Valve / Hydrant

Meter

Water Quality Domain

Water Sample

Quality Parameter (pH, Turbidity, Cl₂...)

Test Result

Compliance Record

Alert / Exceedance Event

Operations Domain

Work Order

Service Request

Supply Interruption Event

Leak / Burst Incident

Maintenance Record

Metering & Consumption Domain

Meter Reading

Consumption Profile

Flow Data (AMI)

Demand Forecast

NRW Calculation

Financial Domain

Bill

Payment

Invoice

Revenue

Budget

Regulatory Domain

License / Permit

Regulatory Filing

Compliance Report

Tariff & Rate Schedule

Audit Record

Geospatial Domain

Network Layer (GIS)

Pressure Zone

District Metered Area (DMA)

Catchment Area

Service Territory Boundary

Application Landscape

Customer Systems

Customer Information System (CIS)

Customer Portal (Web & Mobile)

CRM Platform

IVR & Contact Centre System

Digital Self-Service App

Metering & Billing

Advanced Metering Infrastructure (AMI)

Meter Data Management (MDM)

Billing Engine

Payment Gateway

Collections Management

Water Treatment & Quality

SCADA / Process Control System (WTP)

Laboratory Information Management (LIMS)

Chemical Dosing Control System

Quality Compliance Reporting

Network Operations

Network Management System (NMS)

Pressure & Hydraulic Modelling

Leakage Detection System

Supply Interruption Management

Telemetry & Remote Monitoring

Asset Management

Enterprise Asset Management (EAM)

Geographic Information System (GIS)

Asset Performance Management (APM)

Work & Field Management System

Capital Project Management

Wastewater Operations

WWTP SCADA / Control System

Sewer Network Management

Biosolids Management System

Effluent Compliance Monitoring

Enterprise Systems

Enterprise Resource Planning (ERP)

Human Capital Management (HCM)

Financial Management System

Supply Chain & Procurement

Document Management System

Analytics & Intelligence

Data Warehouse / Data Lake

Business Intelligence Platform

Hydraulic Digital Twin

Predictive Maintenance (AI/ML)

Water Demand Forecasting

Integration & Governance

Enterprise Service Bus (ESB)

API Management Platform

Master Data Management (MDM)

Data Governance Platform

Technology Stack

Compute Infrastructure

- On-Premise Data Centres
- Cloud Infrastructure (AWS / Azure / GCP)
- Hybrid Cloud Architecture
- Virtual Machine Platform
- Container Orchestration (Kubernetes)

Network & Connectivity

- Wide Area Network (WAN)
- Local Area Network (LAN)
- Software-Defined Networking
- MPLS / Leased Lines
- 4G/5G/LTE for Field Operations
- LoRaWAN / NB-IoT for AMI & Sensors

OT / SCADA / ICS

- SCADA (Treatment & Distribution)

- Programmable Logic Controllers (PLCs)
- Remote Terminal Units (RTUs)
- Historian / Time-Series Database
- Telemetry & Remote I/O
- IEC 61850 / DNP3 Protocols

Data Management

- Relational Databases (Oracle, SQL Server)
- NoSQL Databases (MongoDB, InfluxDB)
- Data Lakes (Hadoop, Azure Data Lake)
- Data Warehouses (Snowflake, Synapse)
- Real-time Streaming (Kafka, MQTT)

Security

- Identity & Access Management (IAM)
- Multi-Factor Authentication (MFA)
- Firewalls & IDS/IPS
- SIEM Platform
- Encryption (Data at Rest & in Transit)
- OT / ICS Cybersecurity (Purdue Model)

Integration & Middleware

- Enterprise Service Bus
- API Gateway
- Message Queues (RabbitMQ, Kafka)
- ETL / ELT Tools
- iPaaS (Integration Platform as a Service)

Field Technology

- Smart Water Meters (AMI)
- Pressure Sensors & Data Loggers
- IoT Water Quality Sensors
- Field Mobile Devices & Tablets
- GPS & GIS Mapping Tools
- Acoustic Leak Detection Devices

Disaster Recovery & Backup

- Backup & Recovery Solutions
- Disaster Recovery Site
- High Availability Clustering
- Business Continuity Systems
- Emergency Response Protocols

Security & Privacy

Cybersecurity, OT/IT Security, Data Protection

Water Quality Assurance

Continuous Monitoring, Lab Testing, Standards Compliance

Governance

Architecture Board, Standards, Policies

Regulatory Compliance

Drinking Water Standards, Environmental Licenses, Health Authority

Data Quality & MDM

Master Data, Governance Rules, Quality Assurance

Change Management

Architecture Framework Legend

Strategy Layer: Vision, goals, and strategic objectives

Business Layer: Capabilities, processes, and organization

Information Layer: Data domains, entities, and flows

Application Layer: Systems and applications portfolio

Technology Layer: Infrastructure, platforms, and tools

Cross-Cutting: Security, governance, and compliance

Framework Alignment: This Enterprise Architecture is aligned with TOGAF (The Open

Group Architecture Framework) and follows industry best practices for water utility

operations. It provides a comprehensive view of the organization's strategy, business,

information, applications, and technology in an integrated manner — covering potable

water production, distribution, customer management, wastewater treatment, and NRW

control.